SIGMA56

Causal Reliability-Gated Lead-Lag Ensemble

A theoretical, mathematical, and validation-oriented dissection

Prepared from the submitted Python implementation

Academic explanatory report - source-code interpretation, not an investment recommendation

How to read this report

This document explains Sigma56 at three levels. Section 1 gives a theoretical account of every active strategy and of the economic behaviour the model is attempting to exploit, including the mechanism that is new relative to its immediate predecessor. Section 2 reconstructs those strategies mathematically and shows how their votes are transformed into positions. Section 3 explains the interaction between the primary engine, the independent relationship-decay guard, and the new reliability-gated architecture selector. Section 4 lays out a validation programme for each component and for the integrated model.

> ***Jargon note.*** *"Alpha" means a repeatable source of expected profit that is not merely compensation for broad market exposure. In this report, alpha is discussed as a modelling hypothesis. The source code alone cannot prove that the hypothesis is profitable out of sample.*

Scope and evidential boundary

The report is derived from the uploaded Sigma56 source, file name **Sigma56_CausalReliabilitySleeve.py**. The implementation contains an internal comment header identifying the file as "Sigma54 Causal Reliability + Conservative Sleeve," describing itself as "Sigma51 with a causal three-state selector across the purged cross-fitted one-day LL committee, the frozen Sigma51 forecast, and a conservative 30/40% accumulated-source LL committee." This report treats the file as Sigma56, matching its place in the project's model list, and flags the internal naming discrepancy explicitly rather than silently resolving it. As with its predecessor, the implementation contains internal statistical filters and trailing selectors, but it does not contain a complete historical research log, external performance table, or proof of out-of-sample robustness. Accordingly, the validation chapter distinguishes: (i) validation that is literally implemented inside the model; (ii) validation implied by the architecture; and (iii) external tests that should be run before relying on the strategy.

Primary source: Sigma56_CausalReliabilitySleeve.py. Function and parameter names are cited throughout so the analysis can be checked against the implementation.

Executive summary

Sigma56 is, mechanically, the Sigma51 chassis unchanged in almost every respect: the same seven seed pairs, the same dynamically scanned gray pairs, the same spectrally shrunk two-factor residual lead-lag network, the same ALGO-shock response vector, the same eight-peer ridge basket regression, the same adaptive plain-versus-relative ALGO reversal selector, and the same independent relationship-decay guard. Two things are genuinely new. First, the asymmetric residual-tail reversal sleeve has been retuned to a more conservative configuration: its positive-tail weight is reduced from 0.45 to 0.30 and its activation gate is tightened from two standard deviations to 1.75. Second, and far more materially, the model now embeds a causal reliability sleeve that runs a purged, cross-fitted, walk-forward validation test on the lead-lag structure itself, at every rescan, and uses the result to select among three competing definitions of the lead-lag vote: a fast purged cross-fitted one-day committee, a slower accumulated-source blend, or the untouched Sigma51 forecast.

This is a qualitatively different kind of addition from Sigma51's other sleeves. Every Sigma51 component forecasts price behaviour. The new sleeve instead forecasts *whether the model's own core forecasting mechanism*

is currently trustworthy, using an embedded out-of-sample test, and then acts on that judgement. It is the same philosophical move as Sigma51's adaptive ALGO selector — decide between competing definitions using only causally available evidence — but applied to the entire non-ALGO lead-lag engine rather than to one instrument's reversal definition.

The central modelling idea therefore expands from Sigma51's “diversify across mechanisms and timescales” to add a third axis: “diversify across, and adjudicate between, competing estimators of the same mechanism, using embedded evidence of which estimator is currently replicating out of sample.” Whether this additional layer of adaptivity earns its complexity is precisely the question Section 4 is built to help answer.

Architecture at a glance

Layer	Active mechanism	Intent	Output
Data transformation	Log prices, returns, ALGO residualisation, second-factor removal, cross-sectional standardisation	Strip broad common variation and create comparable residual series	Residual levels and z-scored residual returns
Relationship reversion	Seven seed pairs plus dynamically scanned gray pairs	Trade deviations from persistent relative-value relationships	Dollar votes vR
Temporal network (base)	Spectrally shrunk full and recent lead-lag matrices, identical to Sigma51	Forecast which names tend to move after other names	Base signal used by all three architecture modes
Conditional shock response	AG vector activated after |ALGO z| > 1	Model cross-sectional reactions to exceptional ALGO shocks	Adjustment inside the lead-lag signal
Causal reliability sleeve	Purged, cross-fitted k-fold replication test of the lead-lag matrix, run at every rescan	Measure whether the low-rank propagation structure actually holds out of sample right now	A reliability scalar in roughly [0, 1+]
Architecture selector	9-combination committee vote over reliability low/high thresholds	Choose which of three lead-lag definitions to trust this rescan	Architecture mode 0, 1, or 2
Basket equilibrium	Eight-peer ridge regression of residual levels, identical to Sigma51	Fade a name's deviation from a multivariate equilibrium	Basket vote
Extreme residual reversal	Asymmetric positive/negative tail reversal, retuned more conservatively	Fade one-day idiosyncratic overshoot	Reversal vote
ALGO engine	Adaptive plain vs market-relative reversal, identical to Sigma51	Choose the better recent definition of ALGO mean reversion	±$100,000 ALGO position
Decay guard	Independent stale-relationship committee, identical architecture to Sigma51 with its own spectral constants	Flatten, preserve, or reroute decayed non-ALGO trades	Selective override of primary positions

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1. Theoretical explanation of Sigma56

1.1 Relationship to Sigma51: what is unchanged and what is new

Sigma56 is best read as Sigma51 plus one new decision layer, not as a redesign. Everything Sigma51's dissection established about the primary engine's seed pairs, gray pairs, base spectral lead-lag network, AG-shock vector, basket regression, adaptive ALGO selector, and independent decay guard applies to Sigma56 without modification: the parameter values, thresholds, and formulas for those components are identical between the two files. Two changes distinguish Sigma56 from Sigma51.

- The asymmetric residual-tail reversal sleeve is retuned: the positive-tail vote coefficient falls from 0.45 to 0.30 and the activation gate falls from 2.0 standard deviations to 1.75. The sleeve fires more often (lower gate) but with a smaller maximum vote (lower coefficient) — a strictly more conservative posture on both axes at once, consistent with the file's own header calling itself a “Conservative Sleeve.”
- A new causal reliability sleeve is added around the lead-lag vote specifically. At every twenty-day rescan, the model runs a purged, cross-fitted, walk-forward replication test of the low-rank lead-lag structure and distils the result into a single reliability scalar. A nine-combination threshold committee then selects one of three lead-lag definitions for that rescan: a purged cross-fitted one-day committee (used when reliability is high), a slower accumulated-source blend (used when reliability is low), or the untouched Sigma51 forecast (used in between). Nothing else in the model, including the basket sleeve, the residual-tail sleeve, the ALGO engine, and the decay guard, is touched by this selector.

The remainder of Section 1 restates the shared mechanisms at the same level of detail as the Sigma51 report, so that this document is self-contained, and then devotes 1.8, 1.11, and 1.12 to what is actually different.

1.2 Economic setting and model philosophy

Sigma56 is designed for the same constrained cross-sectional trading environment as Sigma51: one instrument, ALGO, can carry up to $100,000 of exposure, and every other instrument is capped at $10,000. ALGO is traded by its own adaptive reversal engine; the remaining names are traded as a diversified panel of smaller, mostly full-sign positions.

The model remains a committee of hypotheses. Sigma51 asked whether two price levels are temporarily displaced from a stable relative relationship, whether one name tends to lead another by a day, whether an exceptional ALGO shock produces a predictable cross-sectional response, whether a name has drifted from a multivariate peer equilibrium, whether the latest idiosyncratic return is extreme enough to revert, and whether an existing position's supporting relationship has decayed. Sigma56 adds a seventh question, asked not about any single name but about the lead-lag mechanism as a whole: has the low-rank propagation structure that the lead-lag vote depends on actually continued to replicate on data the matrix was not fitted on?

> ***Jargon note.*** *A "committee" is an ensemble in which multiple models contribute to a final decision. Sigma56 contains two nested committees: the original decay-guard committee that votes on individual positions, and a new architecture committee that votes on which lead-lag definition the whole non-ALGO portfolio should use this rescan. Confusing the two is easy but important to avoid, since they operate on different objects and at different points in the pipeline.*

1.3 Preprocessing: why Sigma56 removes common factors

Preprocessing is unchanged from Sigma51. Sigma56 works with log prices and residual returns. Each instrument's return is first regressed on ALGO's return and the ALGO-related component removed; a principal-component extraction on the remaining non-ALGO residual returns then removes the dominant residual common factor; each resulting series is standardised name by name. The same two-stage residualisation also uses a level-based variant, in which each log-price series is centred and regressed on ALGO's centred log price, for use by the pair, basket, and guard mechanisms.

This transformation is foundational to every downstream sleeve, including the new reliability test: the purged cross-fitting machinery is run entirely on the same standardised residual z panel that feeds the base lead-lag matrix, so the reliability sleeve is asking whether structure in that specific residual coordinate system replicates out of sample, not whether raw or differently-processed returns do.

1.4 Seed-pair and gray-pair mean reversion

The seed sleeve contains the same seven predetermined pairs as Sigma51: (10, 46), (1, 20), (13, 45), (8, 27), (49, 50), (25, 37), and (7, 40), indexed against the instrument panel with ALGO at index 0. At each rescan, a seed pair is retained only if its log-price spread passes a full-history autoregressive test (t-statistic below −2.5) and a recent-250-day recurrence test (t-statistic below −1.0). Once retained, the hedge ratio is refit over the trailing 450 observations, the current spread deviation is standardised using an exponentially weighted volatility estimate with a twenty-day half-life, and the model trades against the deviation once it exceeds a quarter-standard-deviation band.

The gray sleeve searches among names not already claimed by a seed pair. For each candidate pair, it regresses one centred log-price series on another, computes the residual spread, and requires a strongly negative autoregressive t-statistic (below −3.4) both over the full history and, in a recent-250-day recurrence check, below −2.8. It also rejects hedge ratios larger than four in magnitude. Qualifying pairs are traded at half weight relative to the seed sleeve, the same confidence discount Sigma51 applied. Both sleeves are entirely unchanged from Sigma51: same thresholds, same windows, same half-weight discount.

1.5 Spectral two-factor residual lead-lag network (the base signal)

The base lead-lag engine is unchanged from Sigma51 and remains the input that every architecture mode either uses directly, blends with an alternative, or ignores. After factor residualisation, it estimates a matrix whose entry from source i to target j measures whether residual return in i at time t is associated with residual return in j at time t+1, with the diagonal and all ALGO rows and columns forced to zero. Two matrices are estimated at every twenty-day rescan: an expanding-history matrix and a 450-day recent matrix, the latter additionally thresholded so raw coefficients below 0.04 are removed. Both matrices are denoised by singular-value shrinkage — scale 1.6 and

power 0.7 for the full-history matrix, scale 3.0 and power 0.95 for the recent matrix — then blended ninety percent full-history and ten percent recent-history. For the full-history matrix only, a small fraction (15%) of unusually large residual edges is restored after low-rank reconstruction, using row and column median-absolute-deviation scaling.

Sigma56 additionally fits a second lead-lag matrix at every rescan, called the aggregate matrix, using exactly the same estimation and shrinkage procedure but built from ten-day trailing averages of the residual z-panel as the source series rather than single-day values. This aggregate matrix plays no role in Sigma51 and exists in Sigma56 purely to support the accumulated-source lead-lag definition described in 1.12; it is fit unconditionally at every rescan regardless of which architecture mode is ultimately selected.

1.6 ALGO-shock response vector

Unchanged from Sigma51. The AG vector modifies the lead-lag forecast only after an exceptional contemporaneous ALGO return. At each rescan, the model estimates how each non-ALGO residual return has historically co-moved with the standardised ALGO return one step earlier. When the latest ALGO z-score exceeds one in absolute value, the estimated response vector, blended ninety percent current and ten percent previous rescan, is multiplied by the signed ALGO shock and added to the lead-lag forecast before that forecast is normalised. This adjustment is shared by every architecture mode: it is applied to the base signal before any mode-specific blending or committee logic runs.

1.7 Multivariate basket residual reversion

Unchanged from Sigma51. For each non-ALGO name, Sigma56 selects the eight other residual-level series whose residual returns are most correlated with it over a recent 250-day window, fits a ridge regression predicting the target residual level from those eight peer residual levels, and reverses the standardised deviation from the fitted basket equilibrium. Only deviations whose standardised score exceeds 1.875 in absolute value are traded, and the vote is capped at forty percent of a standard $10,000 vote at maximum normalised magnitude.

1.8 Asymmetric residual-tail reversal, retuned

This is the first of the two genuine changes from Sigma51. Sigma56 still fades the latest factor-residual return directly, and only in cross-sectional tails, using the same sign-asymmetric structure Sigma51 introduced: the model is more willing to buy names after unusually negative residual moves than to short names after unusually positive residual moves. What changed is the strength and the trigger. Sigma51's positive-tail weight of 0.45 falls to 0.30 in Sigma56, and the activation gate falls from two standard deviations to 1.75. The negative-tail weight, 0.15, is unchanged.

The net effect is not simply “more conservative” in every sense: the lower gate means the sleeve activates on a wider set of days than Sigma51's, capturing tail events that Sigma51 would have ignored as insufficiently extreme, while the lower weight means each activation contributes a smaller vote once it fires. Whether the combination is more or less aggressive in net dollar-vote terms depends on the empirical distribution of tail events between 1.75 and 2.0 standard deviations, which this report's validation chapter treats as an open empirical question rather than a settled design choice.

1.9 Adaptive ALGO reversal selector

Unchanged from Sigma51. ALGO is traded separately at the $100,000 limit. The plain signal reverses the sign of ALGO's five-day log-price move. The relative signal first estimates ALGO's beta to the equal-weight non-ALGO basket over sixty observations, subtracts that basket-related component from ALGO returns, sums the last three residual returns, and reverses the sign of that cumulative idiosyncratic move. The selector computes the trailing, commission-adjusted net-P&L advantage of the relative strategy over the plain strategy across three windows of 280, 300, and 320 days, and selects the relative signal only if it shows a positive trailing edge in at least two of the three windows.

1.10 Relationship-decay guard

Unchanged in architecture from Sigma51, with its own independent parameter set. The guard is a second, independent relationship model, using the same seed pairs and the same gray-pair discovery logic as the primary engine, but with its own spectral shrinkage constants (full-history scale 1.75 and power 0.7, recent scale 2.5 and power 0.75, versus the primary engine's 1.6/0.7 and 3.0/0.95) and its own diagonal multiplier of 1.2 versus the primary engine's 1.0. It carries no basket or residual-tail sleeves and is not affected by the causal reliability selector: the guard always evaluates decay against its own, single, always-on lead-lag matrix, regardless of which architecture mode the primary engine is using that rescan.

For each current non-ALGO position, the guard measures total lead-lag support (including the AG adjustment), base lead-lag support (excluding it), fresh lead-lag support over 75, 90, and 105 days, residual-level support over 200, 225, and 250 days, and an autoregressive stationarity statistic for each residual-level window. A twenty-seven-combination threshold committee votes on whether the position's supporting relationship has structurally decayed; if at least fourteen of the twenty-seven combinations agree, the position is provisionally marked for flattening. A second, independent attribution test then checks whether pair/gray reversion or the AG-specific effect still supports the position: if so, it is preserved; if the pair/gray vote instead points the opposite way, the position is rerouted to that direction; otherwise it is flattened. ALGO is never altered by the guard.

1.11 The causal reliability sleeve

This is the second, and structurally larger, change from Sigma51. At every twenty-day rescan, Sigma56 runs an internal walk-forward validation experiment on its own residual z-panel before deciding which lead-lag definition to trust for the following twenty days.

The experiment works by purged k-fold cross-fitting, a technique built to avoid the two classic ways an in-sample fit can look better than it will perform out of sample: using future information to help fit the model (leakage), and letting the immediate temporal neighbours of a test day contaminate the training set through the same one-day serial structure the model is trying to exploit (adjacency bias). For a chosen number of folds (3, 4, or 5) and a chosen lookback width (375 or 400 days), Sigma56 splits the day-pairs in the window into contiguous 25-day blocks and assigns each block to a fold. It then builds, for each fold in turn, a training lead-lag matrix from every block *except* that fold and a two-day purge zone around it, and a validation matrix from the held-out fold alone. It takes the singular-value decomposition of the training matrix, and asks a direct question of the validation matrix: does the propagation pattern identified on the training data still point the same way, with the same strength, on data that pattern never saw? Only patterns that replicate with the correct sign survive; patterns that reverse sign out of sample contribute nothing.

This same purge/train/validate machinery is used twice, for two different purposes. First, it is used diagnostically: the total amount of positive, sign-confirmed singular mass, averaged across fold counts, lookback widths, and two different ways of assigning day-pairs to folds, becomes a single scalar reliability estimate for how much of the current low-rank lead-lag structure is actually holding up out of sample right now. Second, it is used constructively: the same held-out-confirmed singular strengths, rather than the in-sample training strengths, are used to rebuild an actual forecasting matrix, which becomes the basis of the fast, purged, cross-fitted lead-lag vote available when reliability is judged high.

The diagnostic version of the test is run on a slower, ten-day-aggregated version of the residual z-panel rather than the single-day panel used for the forecasting matrix. This means the reliability scalar is not answering “does tomorrow's fast one-day lead-lag pattern replicate,” but “does the smoothed, lower-frequency version of the same propagation network replicate,” which the model then treats as a proxy for how trustworthy the general low-rank structure is at this point in the data, before deciding how (or whether) to lean on either a fast or a slow lead-lag estimator.

> ***Jargon note.*** *“Purged” and “embargoed” cross-validation are standard techniques from the financial machine-learning literature for handling serially dependent data, where an ordinary random train/test split leaks information because adjacent time periods are correlated. Sigma56's two-day purge zone around each held-out block is a direct, if modest, implementation of that idea. A wider purge is more conservative but discards more training data; the two-day choice here has not been documented as swept, and Section 4 treats its width as one of the parameters requiring an external neighbourhood test.*

1.12 The three-state architecture selector

The reliability scalar produced in 1.11 is converted into a discrete choice by a small threshold committee, structurally similar to the decay guard's committee but operating on the whole non-ALGO lead-lag mechanism rather than on individual positions. Nine combinations of a low threshold (drawn from 0.60, 0.63, 0.66) and a high threshold (drawn from 0.78, 0.80, 0.82) each cast one vote: reliability at or above the high threshold votes for the fast, purged, cross-fitted lead-lag definition; reliability at or below the low threshold votes for the slower accumulated-source blend; anything in between votes for the untouched, frozen Sigma51 forecast. The mode with the most votes among the nine is used for the coming twenty days, until the next rescan re-runs the whole test.

The three modes are not equally strict about position sizing. When the cross-fitted mode wins, the final direction for each name is the median of three signed votes, computed at three different strengths of cross-fitted-forecast influence, which always resolves to a clean full-size long, short, or flat position. When the frozen Sigma51 mode wins, the same clean full-size resolution applies, because it is exactly Sigma51's original sign rule. When the accumulated-source mode wins, however, the final direction is the *average* of two signed votes computed at two different accumulated-blend strengths (30% and 40%); if those two votes disagree in direction for a given name, the result is a half-size position rather than a flat or full one. This is a genuine, if narrow, departure from Sigma51's otherwise strict full-sign-or-nothing positioning discipline, and it is confined entirely to the low-reliability regime.

The selector's own header comment describes it as using “only purged held-out aggregate-source reliability available at each rescan,” with “no absolute calendar boundary and no midpoint reset.” Both claims check out against the implementation: the reliability calculation is built entirely from the price history available at the current decision day, with no reference to a fixed date or a fixed split point in the overall backtest, so the same rule would

fire identically regardless of where in a longer history the current day happens to fall. This is a causally sound design choice, but it is not the same thing as evidence that the choice improves performance; Section 4 separates the two explicitly.

1.13 Overall ensemble behaviour

For each non-ALGO name, Sigma56 forms the same continuous dollar-vote sum Sigma51 used for its non-lead-lag components: relationship reversion plus basket reversion plus residual-tail reversal. The lead-lag contribution is then added on top of this base sum, but which lead-lag definition is added, and with what committee logic around it, depends on the architecture mode selected for the current rescan. ALGO continues to be assigned independently by the adaptive reversal selector, untouched by any of this. After the primary portfolio is built, the independent guard may selectively flatten, preserve, or reroute non-ALGO names exactly as in Sigma51, using its own always-on lead-lag matrix rather than whichever mode the primary engine is currently using.

The practical consequence is that Sigma56's non-ALGO portfolio can, at different points in its history, be running what is functionally three different models under one name: a purged-cross-fitted-forecast model, a slow-accumulated-signal model with occasional half-size positions, or plain Sigma51. The architecture selector is the mechanism deciding, without human intervention and using only causal evidence, which of the three the portfolio is currently entitled to trust.

2. Mathematical breakdown of each strategy

2.1 Notation

Let $P_{i,t}$ be the price of instrument i at decision time t, where i=0 denotes ALGO and i=1,…,N−1 denotes the remaining instruments (N=50). Define log price and one-period log return as

$$p_{i,t} = \log(P_{i,t}), \quad r_{i,t} = p_{i,t} - p_{i,t-1}.$$

Vectors are written in bold and matrices in uppercase. The source-target convention for every lead-lag matrix is chosen so that a forecast is computed by multiplying the current residual row vector by the matrix. All quantities and formulas in 2.1 through 2.8 are identical to Sigma51; they are restated so this report is self-contained.

2.2 Residualisation, pair tests, and pair votes

Log-price levels are centred over the available history, and ALGO's loading is removed:

$$\tilde{p}_{i,t} = p_{i,t} - (1/T)\Sigma_s p_{i,s}, \quad \beta_i^{L} = (\Sigma_t \tilde{p}_{i,t}\tilde{p}_{0,t}) / (\Sigma_t \tilde{p}_{0,t}^{2}), \quad e_{i,t}^{L} = \tilde{p}_{i,t} - \beta_i^{L}\tilde{p}_{0,t}.$$

For a candidate spread x_t, the AR(1) recurrence test estimates $\hat{\phi} = (\Sigma_t x_t x_{t-1})/(\Sigma_t x_{t-1}^{2})$, an approximate standard error $se(\hat{\phi}) = \sqrt{[(1-\hat{\phi}^{2})/(T-1)]}$, and the statistic $t_\phi = (\hat{\phi}-1)/se(\hat{\phi})$. A strongly negative t_ϕ is treated as evidence of mean-reverting rather than unit-root-like behaviour; this is a simplified stationarity screen, not a full augmented Dickey-Fuller test.

For seed pair (i,j), the 450-day hedge relationship $p_{i,t} \approx \alpha_{ij} + \beta_{ij}p_{j,t}$ defines the spread $s_{ij,t} = p_{i,t} - \beta_{ij}p_{j,t} - \alpha_{ij}$. Spread scale is estimated from first differences using an EWMA with half-life 20: $\lambda = 1-0.5^{1/20}$, $v_t = (1-\lambda)v_{t-1} + \lambda(\Delta s_t)^2$, mapped to a level-spread scale by a fixed divisor to give $z^s = (s_T - \text{mean}(s))/\hat{\sigma}_s$. The vote is q=0 if $|z^s| \le 0.25$, otherwise $q = -\text{sign}(z^s)$, and dollar votes are $v_{R,i}$ += 10,000q, $v_{R,j}$ −= 10,000β_{ij}q. Gray pairs use the same spread logic with the ordinary full-history standard deviation and a half-weight discount GW=0.5 before being added into v_R.

2.3 Spectral shrinkage and the base LL forecast

Let Z^- contain residual z-vectors for t=1,…,T−1 and Z^+ contain z_{t+1}. The raw full-history estimator is $L_{raw} = (Z^- Z^{+T})/(T-1)$, with the diagonal and all ALGO rows/columns set to zero. A recent estimator is calculated analogously over the trailing 450 observations, with entries below 0.04 removed. For the non-ALGO block, singular-value decomposition $L_{raw} = U\,\text{diag}(s)V^T$ gives a shrinkage threshold $\lambda = c \cdot \text{median}(s)$ and a retention weight $w(s) = s^p/(s^p+\lambda^p)$. The full-history matrix uses c=1.6, p=0.7; the recent matrix uses c=3.0, p=0.95. For the full matrix only, 15% of unusually large residual edges (relative to robust row/column median-absolute-deviation scales, at a 2.5 threshold) are restored after low-rank reconstruction. The final base matrix is $L = 0.9\,L_{full} + 0.1\,L_{recent}$, and the raw forecast is $g = z_T^{T}L$.

Sigma56 additionally fits an aggregate matrix L_{agg} at every rescan by the identical procedure, but with the source series replaced by a trailing ten-day mean of the residual z-panel: for aggregated source $a_{i,t} = (1/10)\Sigma_{k=0}^{9} z_{i,t-k}$ and target $z_{i,t+1}$, $L_{agg,raw}$ is estimated, shrunk, and blended full/recent by the same constants as L. This matrix plays no role unless the accumulated-source architecture mode is selected (2.13).

Before the AG adjustment, σ_{LL} is the cross-sectional standard deviation of non-ALGO entries of g. After the AG adjustment, define $\sigma_{combined}$ analogously and use $\sigma_{signal} = 0.9\sigma_{combined} + 0.1\sigma_{LL}$. The normalised forecast is $h = g/\sigma_{signal}$, and the base lead-lag vote is $v_{L,i} = 1.3 \times 10{,}000 \times \text{clip}(h_i,-2,2)/2$.

2.4 AG conditional adjustment

Let a_t be the standardised, centred ALGO return. At each rescan, $\gamma_i = (1/(T-1))\sum_t z_{i,t+1}a_t$. When $|a_T|>1$, the lead-lag forecast g is adjusted as $g_i \leftarrow g_i + 2a_T\tilde{\gamma}_i$, where $\tilde{\gamma}$ is a 90/10 blend of the current and previous rescan's estimate. This adjustment is applied to whichever base forecast (single-day g, or the 0.70/0.30 or 0.60/0.40 blends used in the accumulated mode) is being built at the time, since it happens inside the shared vote-construction routine used by every architecture mode.

2.5 Basket regression

For target i, choose eight peers with the highest residual-return correlation over the 250-day basket window. Let X be their residual levels and y the target residual level. After centring, ridge coefficients are $\hat{b} = (X^TX + 0.1\tau I)^{-1}X^Ty$, $\tau = \text{trace}(X^TX)/8$, giving a fitted level $\hat{y}_T = \hat{\alpha} + x_T^T\hat{b}$. The basket reversal score is $z_{B,i} = -(y_T - \hat{y}_T)/\text{sd}(e)$, and after cross-sectional re-standardisation the vote is $v_{B,i} = 0.4 \times 10{,}000 \times \text{clip}(z_{B,i},-2,2)/2$ when $|z_{B,i}| \geq 1.875$, and zero otherwise.

2.6 Tail reversal

Define $z_{R,i} = -z_{i,T}$, re-standardised across names, with clipped shape $c_i = \text{clip}(z_{R,i},-2,2)/2$. Sigma56's retuned coefficients are

$$v_{Rev,i} = 0.30 \times 10{,}000 \times c_i \text{ if } z_{R,i} \geq 1.75; \quad v_{Rev,i} = 0.15 \times 10{,}000 \times c_i \text{ if } z_{R,i} \leq -1.75; \quad v_{Rev,i} = 0 \text{ otherwise.}$$

Sigma51's corresponding constants were a positive-tail coefficient of 0.45 and a symmetric gate of 2.0. Only the positive-tail coefficient and the gate changed; the negative-tail coefficient of 0.15 and the underlying z-construction are identical.

2.7 Adaptive ALGO selector and transaction costs

Unchanged from Sigma51. The plain signal is $q_{plain,t} = -\text{sign}(p_{0,t} - p_{0,t-5})$. Let b_t be the equal-weight mean return of the 49 non-ALGO names; over the trailing 60 observations, $\beta_A = (\sum r_{0,t}b_t)/(\sum b_t^2)$, giving the ALGO-relative residual $\eta_t = r_{0,t} - \beta_A b_t$ and relative signal $q_{rel,t} = -\text{sign}(\eta_t + \eta_{t-1} + \eta_{t-2})$. For each window $W \in \{280,300,320\}$, daily net P&L for either signal is $\pi_t(q) = n_t(q)(P_{0,t+1} - P_{0,t}) - cP_{0,t}|n_t(q) - n_{t-1}(q)|$, c=0.00002, and the trailing edge is $E_W = \sum_t[\pi_t(q_{rel}) - \pi_t(q_{plain})]$. The selected signal is q_{rel} if at least two of E_{280}, E_{300}, E_{320} are positive, otherwise q_{plain}; the dollar position is $D_{ALGO} = 100{,}000q_{ALGO}$.

2.8 Guard committee mathematics

Unchanged from Sigma51. Let $s_i = \text{sign}(n_i)$ denote the guard's own base-position direction. Total and base LL support are $S_{T,i} = s_i v_{L,total,i}/10{,}000$ and $S_{B,i} = s_i v_{L,base,i}/10{,}000$. For recent windows $w \in \{75,90,105\}$, $S_{R,i}(w) = s_i f_{w,i}$ using an unshrunk recent forecast f_w. Residual levels $X_{i,t} = \sum_{s \leq t} z_{i,s}$ give, for level windows $m \in \{200,225,250\}$, $Z_{L,i}(m)$ and directional support $S_{L,i}(m) = -s_i Z_{L,i}(m)$. A structural vote is cast for every combination satisfying $S_{R,i}(w) \leq c_R$, $S_{L,i}(m) \leq -c_L$, $t_{AR,i}(m) \leq -2.5$ for $c_R \in \{-0.25,-0.5,-0.75\}$ and

$c_L \in \{0.5,0.75,1.0\}$, giving structural-vote count V_i. A support committee then votes on $S_{T,i} \geq a$, $S_{B,i} \geq b$, $V_i \geq k$ over $a \in \{0.4,0.5,0.6\}$, $b \in \{0.1,0.25,0.5\}$, $k \in \{1,3,5\}$; a position is marked for decay intervention when at least 14 of the 27 combinations vote true. Preservation and rerouting then depend on pair/gray support $S_{P,i} = s_i v_{R,i}/10{,}000$ and AG-specific support $S_{AG,i} = s_i(v_{L,total,i} - v_{L,base,i})/10{,}000$, exactly as in Sigma51.

2.9 Purged k-fold cross-fitting mechanics

This and the next three subsections describe the mechanism new to Sigma56. Fix a fold count $K \in \{3,4,5\}$ and a day-pair sequence of length M (either the one-day z-lag pairs used for the forecasting matrices, or the ten-day-aggregated pairs used for the reliability diagnostic). Assign block index $b_t = \text{floor}(t/25)$ to each day-pair and fold label $\phi(t) = b_t \bmod K$. Two labelling conventions are used: a *relative* labelling in which t is counted from the start of the current lookback window, and an *absolute* labelling in which t is counted from a fixed historical origin (nt–M–2 for a window of width M ending at the current day nt), so that a given historical day retains a stable fold membership across rescans even as the window slides forward.

For fold k, the validation set is $V_k = \{t : \phi(t)=k\}$. The purge set dilates V_k by two positions in each direction, and the training set is everything outside the union of V_k and its purge zone. Training and validation lead-lag matrices, L_{train} and L_{valid}, are estimated by the ordinary source-times-target inner-product formula restricted to their respective day-pair sets, with zero diagonals. The SVD $L_{train} = U\Sigma V^T$ gives, for each singular index m, a held-out replication score

$$s_m = u_m^T L_{valid} v_m, \quad \rho_m = \max(s_m, 0).$$

ρ_m is the amount of the m-th latent propagation mode's strength that is confirmed, with the correct sign, on data the mode was not fit on; modes whose held-out projection is negative (the pattern reversed direction out of sample) contribute zero. This calculation is performed once per fold and averaged.

2.10 Aggregate-source reliability statistic

The diagnostic reliability scalar uses ρ_m from 2.9 applied to the ten-day-aggregated source/target pairs. For a fold count K and labelling ϕ, the positive-mass statistic for one fold is $\sum_m \rho_m$, and the fold-averaged statistic is

$$R(K,\phi,w) = (1/K) \sum_{k=1}^{K} \sum_m \rho_m^{(k)}$$

computed for both lookback widths $w \in \{375,400\}$, both labellings $\phi \in$ {relative, absolute}, and all three fold counts $K \in \{3,4,5\}$, giving twelve values. The aggregate reliability used by the architecture selector is their simple mean:

$$R_{avg} = (1/12) \sum_{w,\phi,K} R(K,\phi,w).$$

R_{avg} has no fixed upper bound in principle, since it is a sum of positive held-out singular-value replications rather than a bounded correlation, but in practice it behaves as a small positive number whose typical range is what motivates the 0.60–0.66 and 0.78–0.82 threshold bands used by the selector in 2.11.

2.11 Architecture-mode committee vote

For every combination of a low threshold $lo \in \{0.60,0.63,0.66\}$ and a high threshold $h \in \{0.78,0.80,0.82\}$, nine in total, a vote is cast:

vote(lo,h) = 0 (cross-fitted LL) if $R_{avg} \geq h$; vote(lo,h) = 1 (accumulated-source LL) if $R_{avg} \leq lo$; vote(lo,h) = 2 (frozen Sigma51 forecast) otherwise.

The architecture mode for the coming rescan period is the mode receiving the most votes, $\hat{m} = \text{argmax}_{k \in \{0,1,2\}} |\{(lo,h) : \text{vote}(lo,h)=k\}|$. Because the nine (lo,h) pairs are not independent draws but a small predeclared neighbourhood, this behaves as a robustness check on the exact threshold choice rather than as nine independent pieces of evidence; a reliability estimate that changes mode only under one or two of the nine combinations is a weaker basis for switching than one that is unanimous.

2.12 Cross-fitted LL forecast and the mode-0 vote

When $\hat{m}=0$, the primary engine uses the constructive form of 2.9: for each fold, $(U\cdot(\Sigma\cdot\rho))V^T$ replaces the training reconstruction, averaged across folds to give one matrix per fold count, and averaged across fold counts to give one matrix per lookback width and labelling. Given the current one-day residual z-vector z_T, each matrix produces a normalised forecast $\text{norm}(z_T{}^T M)$; the relative-labelled and absolute-labelled forecasts are each averaged across the two lookback widths and re-normalised, then the two labellings are averaged and re-normalised a final time to give a single cross-fitted forecast f_{CF}. The cross-fit vote is

$$v_{CF,i} = 10{,}000 \times \text{clip}(f_{CF,i}, -2, 2)/2 \text{ if } |f_{CF,i}| \geq 1.15, \text{ else } 0.$$

Let $C = v_R + v_B + v_{Rev}$ denote the shared, non-lead-lag vote sum, and let v_L be the base lead-lag vote of 2.3 (with the AG adjustment of 2.4 applied). For each $\alpha \in \{0.60,0.75,0.90\}$, a committee direction is formed as $d_\alpha = \text{sign}(C + v_L + \alpha v_{CF})$; the mode-0 target direction is the elementwise median of the three, $\text{med}(d_{0.60}, d_{0.75}, d_{0.90})$, which is always exactly one of the three input directions and therefore always in $\{-1,0,1\}$.

2.13 Accumulated-source LL forecast and the mode-1 vote

When $\hat{m}=1$, the primary engine instead uses the aggregate matrix L_{agg} of 2.3. Let $g_{agg} = (\text{mean of the trailing ten-day z-window})^T L_{agg}$ be the aggregate-matrix forecast and g the base one-day forecast of 2.3. Two blended signals are built,

$$g_1 = 0.70g + 0.30g_{agg}, \quad g_2 = 0.60g + 0.40g_{agg},$$

each run through the same normalisation and vote-construction routine as the base signal (including its own AG adjustment) to give $v_L^{(1)}$ and $v_L^{(2)}$. Two directions are then formed, $d_1 = \text{sign}(C + v_L^{(1)})$ and $d_2 = \text{sign}(C + v_L^{(2)})$, and the mode-1 target direction is their simple average, $(d_1+d_2)/2$. Unlike the mode-0 median, this average is not guaranteed to land in $\{-1,0,1\}$: whenever d_1 and d_2 disagree with one of them zero and the other ±1, the target direction is ±0.5, and the eventual position for that name is half of the standard \$10,000 leg. This is the one place in Sigma56 where the discrete, full-sign positioning discipline documented throughout Sigma51 is not strictly preserved.

2.14 Vote aggregation, caps, and state

For each i>0, the primary target is $n_i = d_i \times 10{,}000/P_{i,T}$, using whichever d (mode-0 median, mode-1 average, or plain $\text{sign}(C+v_L)$ in mode 2) the architecture selector chose for the current rescan period. As in Sigma51, ALGO's position is assigned independently by the adaptive selector of 2.7, the optional beta hedge is computed but multiplied by a zero weight, and the guard of 2.8 may override individual non-ALGO names after the primary

portfolio is built. All state — the seed/gray/birth unit lists, both lead-lag matrices, the aggregate matrix, the AG vector, the basket coefficients, the cross-fitted matrices, and the architecture mode itself — is refreshed together at each twenty-day rescan and held fixed in between.

3. Integrated model mechanics and theoretical interpretation

3.1 Why the strategies can coexist

As in Sigma51, the strategies operate on different objects, so two sleeves can disagree without one being mathematically inconsistent. Pair and basket sleeves use residual levels; the lead-lag family (in any of its three architecture-selected forms) uses residual returns across time; the AG vector uses conditional cross-sectional response to ALGO; tail reversal uses the latest residual return; ALGO reversal uses ALGO's own multi-day trajectory; and the guard uses persistence of support for an already-selected direction. Sigma56 resolves conflicts the same way Sigma51 did, by vote magnitude, except that the vote magnitude contributed by the lead-lag family now itself depends on a prior, separately-adjudicated decision about which lead-lag definition to trust.

3.2 Timescale decomposition

Mechanism	Typical information horizon	Economic timescale
Residual-tail reversal	Latest one-day residual	Very short overshoot/liquidity correction, now gated at 1.75σ instead of 2.0σ
Base lead-lag	One-day source-to-target propagation	Short information diffusion
Cross-fitted lead-lag (mode 0)	One-day propagation, filtered to only out-of-sample-replicated modes over 375/400-day lookbacks	Short diffusion, conditioned on demonstrated recent reliability
Accumulated-source lead-lag (mode 1)	Ten-day aggregated propagation, 30% or 40% blended with the base signal	Slower diffusion, used precisely when the fast signal is judged unreliable
AG vector	Immediate response after exceptional ALGO shock	Shock-conditioned short horizon
Adaptive ALGO relative reversal	Three-day idiosyncratic ALGO residual	Short relative overextension
Plain ALGO reversal	Five-day ALGO move	Short momentum exhaustion
Pair reversion	Up to 450-day relationship fit	Medium-horizon equilibrium restoration
Basket reversion	250-day multivariate fit	Medium-horizon manifold convergence
Causal reliability sleeve	375/450-day purged cross-fitted replication test, refreshed every 20 days	Meta-horizon judgement of whether the fast propagation mechanism currently generalises
Decay guard	75-250-day support diagnostics	Relationship life-cycle control

3.3 What the reliability sleeve adds to alpha attribution

Sigma51's dissection argued that factor residualisation helps isolate name-specific propagation from shared market response, but cannot eliminate all confounding, because factor estimates are themselves noisy and nonlinear relationships may remain. The causal reliability sleeve is best read as a direct response to exactly that caveat: rather than assuming the low-rank propagation structure is equally trustworthy at all times, it measures, causally and out of sample, whether it currently is, and adjusts how much the model leans on it. This does not remove the underlying confounding risk that any lead-lag estimate is associational rather than causal in the economic sense, but it does add a live, embedded check against one specific failure mode: a propagation pattern that was real in the fitting window but has since stopped holding.

3.4 Why a three-way selector rather than a single continuous blend

An alternative design would have used the reliability scalar directly as a continuous blend weight between the fast and slow lead-lag definitions, avoiding a hard mode switch. Sigma56 instead makes a discrete choice among three named regimes, using a nine-combination threshold committee to decide between them. This mirrors the same design instinct Sigma51's decay guard already used for individual positions: discrete action driven by a small robustness committee, rather than a single point estimate driving a continuous adjustment. The tradeoff is the same one Sigma51 itself flagged for full-sign positioning: discreteness can reduce sensitivity to small estimation errors away from a boundary, but increases sensitivity to reliability estimates that sit close to one of the two thresholds, where the portfolio's entire lead-lag definition can flip based on a small change in R_{avg}.

3.5 The one place full-sign discipline is relaxed

Section 2.13 noted that mode 1 can produce a half-size position when its two internal blend variants disagree. This is worth dwelling on theoretically, not just mathematically. Sigma51's report argued that full-sign positioning is “both robust and dangerous”: robust because it discards uncertain magnitude estimates, dangerous because it creates a discontinuity where a small change near zero can flip a name from full long to full short. Mode 1's half-size positions are, in effect, a built-in acknowledgement of that same discontinuity risk, but only in the specific regime the model has judged least reliable: when reliability is low, and the two candidate accumulated-blend strengths disagree about direction, the model hedges its own uncertainty by taking half size rather than forcing a full-conviction bet. Whether this is a deliberate design choice or an incidental consequence of averaging two sign vectors is not stated in the source; either way, its practical effect (smaller positions exactly when the model's own diagnostics say it is least sure) is directionally sensible and worth confirming empirically rather than assuming.

3.6 The guard as an unconditional anchor

One structural property is easy to miss and important to state plainly: the decay guard is never told which architecture mode the primary engine is using, and does not change its own behaviour based on it. The guard always evaluates every non-ALGO position against its own single, always-on lead-lag matrix, computed with the guard's own spectral constants, regardless of whether the primary engine's position was generated under mode 0, 1, or 2. This makes the guard a stable outside reference: if the architecture selector begins making poor mode choices, the guard's flatten/preserve/reroute logic is unaffected and continues to check positions against the same yardstick it always has. It also means the guard cannot correct for a systematic bias introduced by the selector itself, since the guard's support measurements do not have visibility into which lead-lag definition actually produced the position it is evaluating.

3.7 State, rescans, and path dependence

As in Sigma51, both engines maintain state and rescan every twenty observations, so relationships, matrices, and now also the architecture mode are piecewise constant between rescans. Sigma56 adds one further layer of path dependence beyond Sigma51's: the architecture mode itself is a piece of state that depends on the exact rescan phase, because the reliability calculation, like everything else, is recomputed only every twenty days from whatever history happens to be available at that point. A cold start or state reset does not merely change which pairs or matrices are active, as in Sigma51; it can also change which of the three lead-lag definitions the model believes it should be using, for the entire following twenty-day block. Validation must reproduce the exact deployment path for this reason even more emphatically than it needed to for Sigma51.

4. Validation framework for every strategy

This chapter separates internal gates from external evidence, exactly as in the Sigma51 report. An internal gate is part of the trading rule; it does not establish that the rule is profitable. External validation must evaluate forecasts and realised net P&L on observations that did not determine the rule, parameter, or model choice. Sections 4.2 through 4.9 restate the Sigma51 protocol for the shared components, since none of their internal logic changed; 4.10 covers the retuned tail-reversal sleeve; 4.11 and 4.12 cover the two genuinely new mechanisms; 4.13 covers the guard; and 4.14 covers the integrated portfolio.

4.1 Universal validation protocol

- Freeze the strategy definition before evaluating a held-out period. Record all candidate variants considered.
- Use causal walk-forward replay, including exact commissions, dollar limits, integer share rounding, warm-up, and state-reset behaviour.
- Report non-overlapping 250-day blocks and the continuous horizon. Do not count overlapping rolling windows as independent evidence.
- Perform component ablation inside the full portfolio and standalone isolation. A sleeve can be weak alone but valuable in combination, or vice versa.
- Use block-level consistency tests, leave-one-block-out tests, and instrument concentration tests.
- Test a small predeclared parameter neighbourhood. A lone optimum surrounded by losses is evidence of selection risk.
- Use structural placebos: reverse the sign, shift timing, transpose directed matrices, shuffle instruments, or replace lag one with lag zero/two.
- Stress commissions, position limits, rescan phase, warm-up length, and missing or perturbed observations.
- Check forecast diagnostics separately from portfolio score: hit rate, calibration, turnover, correlation with existing sleeves, and incremental P&L attribution.
- Document failed variants. The effective number of trials matters when interpreting the best observed result.

4.2 through 4.9: shared components carried over from Sigma51

Section	Required external tests (unchanged from Sigma51)
4.2 Seed-pair validation	Ablate each of the seven pairs one at a time and jointly; freeze pair identities before the test period; test hedge-ratio windows near 450 as a small neighbourhood; run spread-sign inversion and leg-swap controls; test concentration by removing the best pair, best three, and worst pair.
4.3 Gray-pair validation	Separate discovery and evaluation so pairs are scored only after selection; compare against shuffled and random-matched pair sets; measure survival and turnover in membership; treat the 0.5 weight as a broad discount, not a fine-tuned optimum.

Section	Required external tests (unchanged from Sigma51)
4.4 Spectral lead-lag validation	Lag placebos (lag zero, lag two, reversed time, transposed direction); sign placebo (invert the forecast); SVD ablations (raw, low-rank only, sparse only, shuffled/permuted singular structure); test the frozen 90/10 full/recent blend for net value after turnover.
4.5 Sparse edge restoration validation	Ablate the 15% restoration entirely and test nearby strengths; compare to symmetric, antisymmetric, reciprocal-only, and wrong-lag restoration; test whether restored edges add information beyond the low-rank modes.
4.6 AG-shock vector validation	Isolate positive and negative ALGO shocks and test forecast symmetry; run timing placebos (shock one day late/early, vector shuffled); ablate the ordinary LL matrix to see whether AG works independently or only by resolving conflicts.
4.7 Basket residual validation	Peer-selection placebo with random and least-correlated peers; embargo selection and fitting so current error cannot influence peer choice; test nearby peer counts, ridge strengths, and windows as broad structures.
4.8 Adaptive ALGO selector validation	Compare plain and relative signals separately over non-overlapping and continuous periods; test whether the selector beats the best fixed strategy chosen without test data; apply an embargo so the most recent decision is not reused; simulate equal-edge candidates to quantify selection noise.
4.9 Decay guard validation	Compare the complete guard with flatten-only, preserve-only, and reroute-only ablations; run random intervention placebos matched on frequency; shift guard decisions forward/backward in time; report precision (would the flattened position have lost money) and opportunity cost (gains lost by flattening positions that recovered).

4.10 Retuned residual-tail reversal validation

Internal validation already present: factor-adjusted residual input, cross-sectional standardisation, a 1.75-standard-deviation tail gate, clipping, and distinct positive/negative weights, all unchanged in kind from Sigma51 and differing only in the specific constants.

- Because both the gate and the positive-tail coefficient moved at once, decompose the change: test the lowered gate (1.75 instead of 2.0) with Sigma51's original 0.45 weight, and Sigma51's original 2.0 gate with the new 0.30 weight, as two separate intermediate variants, to learn which change is doing the work rather than treating the combination as a single unit.
- Report event counts under the 1.75 gate versus the 2.0 gate specifically for days that fall between 1.75 and 2.0 standard deviations, since these are the only days whose treatment changed at all.
- Re-run the sign-inversion, one-day-delay, and symmetric-weight placebos Sigma51's protocol already specified, using the new constants, since a placebo result under the old constants does not certify the new ones.
- Confirm that the net dollar-vote contribution of the retuned sleeve is not simply smaller in every regime; if it is, the change should be described as a size reduction rather than as a retuned trigger.

4.11 Causal reliability diagnostic validation

Internal validation already present: purging with a two-day embargo around each held-out block, two independent fold-labelling conventions (relative and absolute), three fold counts, two lookback widths, and a floor at zero for held-out singular-value replication so only sign-confirmed modes count. This is a genuine, non-trivial internal out-of-sample check, considerably more rigorous than a simple in-sample diagnostic, but it is still an internal gate: it has not been shown, only asserted by construction, to correlate with which lead-lag definition will actually perform better going forward.

- Directly test the central causal claim: on a held-out period, does R_{avg} measured just before that period actually predict which of the three architecture modes would have performed best during it? This is the single most important test this report can specify, since the entire sleeve's justification rests on that predictive relationship holding.
- Vary the purge width (currently fixed at two days) and the block width (currently fixed at 25 days) as a small neighbourhood; a reliability estimate that is highly sensitive to either choice is weaker evidence than one that is stable across a plausible range.
- Test the two fold-labelling conventions separately rather than only in their averaged form, to learn whether the absolute labelling's cross-rescan stability actually adds value over the simpler relative labelling, or is redundant with it.
- Run a shuffled-instrument placebo on the held-out validation matrix specifically: if names are randomly relabelled only within the validation fold, held-out replication should collapse toward zero; if it does not, the diagnostic may be picking up a structural artefact (for example, common shrinkage direction) rather than genuine name-specific propagation.
- Report the reliability scalar's own time series over the full available history, including its full range, so that the specific 0.60-to-0.66 and 0.78-to-0.82 threshold bands can be checked against where the statistic actually spends its time, rather than assumed to be well-calibrated.

4.12 Architecture-selector and cross-fitted/accumulated LL validation

Internal validation already present: a nine-combination threshold committee analogous in spirit to the decay guard's own committee, a three-alpha robustness sweep for the mode-0 median vote, and a two-blend comparison for the mode-1 average vote.

- Report standalone performance for each of the three modes forced on for the entire test period, separately from the adaptive selector's realised performance. As Sigma51's own ALGO-selector protocol required, adaptive switching must be shown to add value beyond the best single fixed choice made without test-period information, not merely assumed to.
- Report mode-switch frequency, the distribution of time spent in each mode, and portfolio turnover immediately around switches, since a selector that changes the entire non-ALGO lead-lag definition every rescan would impose materially different transaction costs than one that is stable for long stretches.
- Test whether the half-size positions unique to mode 1 (Section 2.13) improve or harm realised P&L relative to a variant that instead forces mode 1's target direction to the full sign of d_1+d_2 whenever either is nonzero, isolating the effect of that specific design choice.

- Ablate the cross-fitted forecast's own gate (currently 1.15) and its three-alpha committee (currently 0.60/0.75/0.90) as a small neighbourhood, using the same lone-optimum-versus-plateau logic Sigma51's protocol applied to every other gate and coefficient in the model.
- Test rescan-phase and cold-start sensitivity specifically for the architecture mode, given the additional path dependence identified in 3.7: repeat the full backtest starting from several different history lengths and confirm the sequence of selected modes, not only the final portfolio score, is reasonably stable.

4.13 Decay guard validation

Unchanged in method from Sigma51's protocol, restated because the guard's own spectral constants differ slightly from the primary engine's in Sigma56 and should therefore be validated in their own right rather than assumed to inherit Sigma51's guard evidence.

- Compare the complete guard with flatten-only, preserve-only, and reroute-only ablations, and with random intervention placebos matched on frequency and instrument distribution.
- Evaluate the independent guard base itself, so guard improvements are not confused with differences between the guard's spectral constants (1.75/0.7 full, 2.5/0.75 recent, 1.2 diagonal multiplier) and the primary engine's (1.6/0.7 full, 3.0/0.95 recent, 1.0 diagonal multiplier).
- Confirm the guard's insensitivity to the primary engine's architecture mode empirically, not only structurally: intervention rates and precision should look similar regardless of which mode the primary engine happened to be running when a given position was opened.

4.14 Integrated portfolio validation

- Report the official score, mean P&L, standard deviation, Sharpe ratio, turnover, and commissions for every horizon, including any newly available continuous window.
- Perform cumulative ablation in a predeclared order: base sleeves only, base sleeves plus base lead-lag only (equivalent to forcing mode 2 permanently, i.e. plain Sigma51), then the full adaptive selector, so the reliability sleeve's marginal contribution over plain Sigma51 is measured directly rather than inferred.
- Report daily P&L correlation between Sigma56 and a frozen Sigma51 run on the same price history, and the fraction of days and names on which the two portfolios actually differ, to quantify how much of the time the new machinery is doing anything at all.
- Test exact state-reset policy, repeated calls at identical history length, shorter-history restart, and rescan phase, with particular attention to the architecture mode's own path dependence noted in 3.7.
- Stress 2x and 3x commissions, half limits, integer rounding, and alternative liquidation treatment at the test endpoint.
- Keep a model-selection ledger and apply a multiple-testing penalty or deflated Sharpe analysis; Sigma56 adds a materially larger number of interacting internal choices (fold counts, lookback widths, labellings, alphas, blend weights, threshold bands) than Sigma51 already had, so this penalty should be larger, not the same, for Sigma56.

5. Parameter map, inactive scaffolding, limitations, and glossary

5.1 Active primary parameters shared with Sigma51

Parameter	Value	Role
WARMUP	120	No positions before sufficient history
RESCAN	20	Refresh relationships, matrices, and the architecture mode every 20 observations
LEG	$10,000	Non-ALGO vote and position cap
WIN	450	Seed-pair regression window
NFAC	2	Remove ALGO plus one dominant residual factor
LL_ROLL	450	Recent lead-lag window (base and aggregate matrices)
LL_BLEND	0.10	Recent matrix share (base and aggregate matrices)
LLW	1.30	Lead-lag vote scale
LL_FULL_SV_SCALE / POWER	1.6 / 0.7	Full-history spectral shrinkage (primary engine)
LL_RECENT_SV_SCALE / POWER	3.0 / 0.95	Recent-history spectral shrinkage (primary engine)
AG_THRESH	1.0	ALGO shock activation threshold
AG_WEIGHT	2.0	AG response strength
BASKET_WINDOW / PEERS / RIDGE / GATE / ALPHA	250 / 8 / 0.1 / 1.875 / 0.4	Basket regression configuration
ALGO selector windows	280, 300, 320	Trailing comparison horizons
ALGO beta window / residual horizon / plain horizon	60 / 3 / 5	ALGO reversal definitions

5.2 Retuned parameters

Parameter	Value	Role
REV_POS_ALPHA	0.30 (Sigma51: 0.45)	Positive reversal-tail weight
REV_GATE	1.75 (Sigma51: 2.0)	Residual-tail activation threshold

5.3 New parameters: causal reliability sleeve and architecture selector

Parameter	Value	Role
CF_LOOKBACKS	375, 400	Lookback widths for cross-fitting
CF_FOLDS	3, 4, 5	Fold counts for purged cross-fitting
CF_BLOCK	25	Block width for fold assignment
CF_PURGE	2	Purge distance around each held-out block
CF_ALPHAS	0.60, 0.75, 0.90	Mode-0 committee weights on the cross-fitted forecast
CF_GATE	1.15	Cross-fitted forecast activation gate
Aggregate blend weights	0.70/0.30 and 0.60/0.40	Mode-1 base/aggregate blend strengths
_ARCH_RELIABILITY_LOWS	0.60, 0.63, 0.66	Low-reliability committee thresholds
_ARCH_RELIABILITY_HIGHS	0.78, 0.80, 0.82	High-reliability committee thresholds

5.4 Guard parameters

Parameter family	Values	Purpose
Recent LL windows	75, 90, 105	Detect current directional support
Residual-level windows	200, 225, 250	Measure long disequilibrium and recurrence
Guard spectral constants (full / recent / diagonal)	1.75/0.7, 2.5/0.75, 1.2	Guard's own, independent shrinkage constants (Sigma51: 1.6/0.7, 3.0/0.95, 1.0)
Committee threshold	14 of 27	Trigger decay intervention
Preserve thresholds	0.25, 0.5, 0.75	Independent pair/AG support tests

5.5 Inactive or dormant features

Unchanged from Sigma51, all still present in the source but inactive in the submitted configuration:

- Birth-pair discovery: BIRTHW=0, so no birth units are created or traded.
- Single-name level reversion: MAX_SNG=0, so candidates are calculated but none are admitted.
- Rank-peer overlay: RANK_PEER_ALPHA=0, so no active rank-peer vote is applied.
- ALGO beta hedge: ALGO_HEDGE_WEIGHT=0, so the calculated hedge dollars do not affect ALGO.
- PCA diagnostics: the module-level _primary_pca_z and _primary_pca_vote variables are declared and reset but never populated or used.
- Seed singles: the list is empty.

These should not be described as active Sigma56 strategies. They are scaffolding or disabled branches, carried over unchanged from Sigma51.

5.6 Principal limitations

- Every limitation documented for Sigma51 applies unchanged: many interacting choices creating substantial multiple-testing risk, simplified AR recurrence screens that do not prove cointegration, linear expanding-history factor residualisation that may leave nonlinear or regime-dependent structure in place, associational rather than causal lead-lag estimates, a discontinuous full-sign portfolio (mostly, see below), basket peer selection and regression refit on the same window, and an ALGO selector that adapts on trailing realised P&L and may chase noise despite its three-window vote.
- Sigma56 adds its own limitations on top. The reliability diagnostic and the cross-fitted forecasting matrices are estimated from heavily overlapping 375-400-day windows that slide forward only twenty days between rescans; successive reliability estimates are therefore highly autocorrelated and should not be treated as independent confirmations of the sleeve's validity.
- The architecture selector adds a third layer of adaptivity on top of two that Sigma51 already had (the ALGO plain/relative selector, and the guard's own flatten/preserve/reroute committee). Each additional adaptive layer increases the space of implicit historical choices the current configuration represents, which increases the burden on external, out-of-sample evidence before the configuration should be trusted.
- The causal soundness of the selector (no lookahead, no fixed calendar boundary) is a necessary property for a walk-forward system, but it is not sufficient evidence that switching lead-lag definitions based on R_{avg} actually improves realised performance; that is an empirical claim this report has not tested and Section 4.11 specifies how it should be tested.
- Mode 1's occasional half-size positions are a narrow but real exception to the full-sign positioning discipline that both Sigma51 and the rest of Sigma56 otherwise maintain, and should be flagged rather than overlooked when reasoning about position sizing or risk.
- The source code does not by itself establish historical performance, statistical significance, or future profitability.

5.7 Amateur glossary

Term	Plain-language meaning
Alpha	Expected return attributable to a repeatable forecasting edge rather than broad market exposure or chance.
Cross-fitting	Fitting a model on one subset of data and evaluating it on a disjoint subset, repeated across several subset splits (folds), so that every observation is eventually evaluated out of sample.
Purging / embargo	Removing observations adjacent in time to a held-out test block from the training set, to prevent serial correlation from leaking information across the train/test boundary.
Held-out singular-value replication	The degree to which a latent propagation pattern (singular vector pair) discovered on training data continues to point the same way, with the same strength, on a disjoint validation set.

Term	Plain-language meaning
Reliability scalar	A single number summarising how much of a model's low-rank structure has been confirmed to replicate out of sample, used here to gate which lead-lag definition is trusted.
Architecture mode	One of three named regimes (cross-fitted, accumulated-source, or frozen Sigma51) that determine how the lead-lag vote is constructed for the current rescan period.
AR(1)	A model in which the current value is related to the previous value. A coefficient below one is consistent with pullback toward a mean.
Factor residual	The part of a return left after subtracting estimated common-factor effects.
Gate / band	A threshold below which a signal is ignored to reduce noise and turnover.
Lead-lag	A relationship in which movement in one instrument tends to precede movement in another.
Low rank	A matrix described by a small number of latent patterns rather than many independent edges.
Ridge regression	Linear regression with an L2 penalty that stabilises coefficients when predictors are correlated.
Sharpe ratio	Mean P&L divided by P&L volatility, commonly annualised. It measures risk-adjusted performance, not certainty.
Singular value	The strength of a latent source-target pattern in a matrix decomposition.
Vote space	The intermediate dollar-like scale in which components are added before the final sign (or, in mode 1, the average of two signs) is selected.
Walk-forward validation	Repeatedly fitting only on past data and evaluating on subsequent unseen data.

5.8 Final interpretation

Sigma56 is Sigma51 with a meta-layer added on top of one mechanism. Every economic hypothesis Sigma51 encoded — equilibrium convergence in pairs and baskets, one-day information diffusion through the lead-lag network, conditional shock propagation through the AG vector, extreme-event reversal, adaptive ALGO overextension, and relationship mortality through the decay guard — is still present in Sigma56, essentially unaltered. What Sigma56 adds is a hypothesis about the lead-lag mechanism's own reliability: that a purged, cross-fitted, walk-forward test of whether the low-rank propagation structure is currently replicating out of sample carries useful information about which definition of that structure — fast and selective, slow and blended, or the original unmodified estimator — the portfolio should use for the next twenty days.

That hypothesis is more sophisticated than anything in Sigma51, and its internal construction is genuinely more rigorous than a plain in-sample fit: the purge/embargo discipline, the dual relative/absolute fold labelling, and the requirement that a mode only be sign-confirmed on held-out data before it counts, are all real methodological improvements over doing nothing. But sophistication of construction is not the same thing as demonstrated value, and Sigma51's own closing argument applies here with, if anything, more force: the more layers a model contains,

the easier it is to explain historical success after the fact. Confidence in Sigma56 relative to Sigma51 should rest specifically on whether the architecture selector's mode choices, tested walk-forward and out of sample, actually outperform simply running Sigma51 unmodified — the single comparison this whole report keeps returning to as the one that matters most.

Source and methodological note

This report interprets the exact implementation supplied as **Sigma56_CausalReliabilitySleeve.py**. The file's own internal header comment names it “Sigma54 Causal Reliability + Conservative Sleeve” and its internal function and variable names use the same _primary_/_guard_ prefixing convention as Sigma51's source; this report follows the project's own model-list naming of the file as Sigma56 throughout, and notes the internal “Sigma54” label here rather than silently reconciling it. Strategy names are descriptive labels assigned to code paths; they are not claims made by an exchange, issuer, or academic paper. Mathematical notation has been standardised for readability, so expressions may differ syntactically from NumPy operations while representing the same computation.

No claim of future profit, statistical significance, or economic causality is made. The validation protocols in Section 4 are designed to test those claims separately.